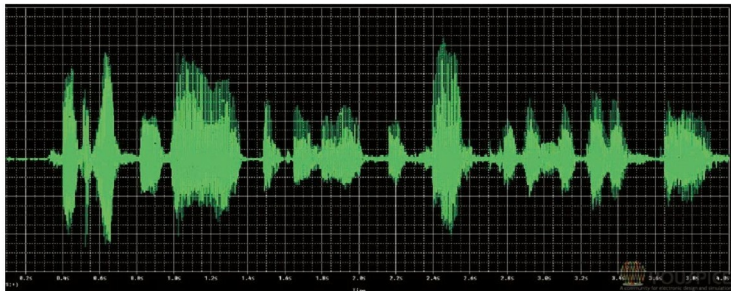


Audacity: Yet Another Tool for Speech Signal Analysis

Most of us have heard of Audacity in connection with making ringtones. But it has many more uses, such as recording live audio from a voice or from musical instruments, creating audio books, mixing music, and so on. This article gives readers a good introduction to the software.



Audacity is a free, multi-track audio editor and recorder. An open source tool for speech analysis, its interface is translated into many languages, and its source code is available from <https://github.com/audacity/audacity>.

Audacity has several functionalities. It is used to:

- Record live audio from voice or musical instruments
- Record audio from the system (e.g., Media Player, YouTube audio, etc)
- Create podcasts by mixing voice with background music
- Deploy audio from any external recording device (tape decks, radio, video cassette recorders, personal video recorders, DVD players, TVs, etc) to the digital format (CD, computer, portable music player, etc)
- Edit sound files with extensions like WAV, AIFF, FLAC, MP2, MP3, etc
- Apply many effects, such as removing vocals from a song, changing the pitch/speed/frequency/amplitude of the audio, generating or removing the noise, etc
- Split an audio into separate tracks
- Export audio files to iTunes
- Create audio books for kids
- Plot the spectrum and spectrogram of a sound file

Installation guide

Audacity is a cross-platform tool, available for Windows, Mac and Linux. Its latest version is Audacity 2.1.2. There are

many ways to get this software:

- From the official Audacity website
- From the Ubuntu Apps Directory (for Linux)
- From the SourceForge website

After downloading the software, the installation process is very simple to follow, as directed by the installation wizard.

Getting familiar with the Audacity environment

Audacity offers various functionalities. Some of them are explored here by using the screenshots from Audacity's official documentation website (Reference 4). Interested readers could refer to this site for more details.

The most commonly used toolbars are described below.

Transport Toolbar: This manages recording and playback.

Tools Toolbar: This is used for audio selection, volume adjustment, zooming and time-shifting.

Recording Meter Toolbar: This displays recording levels.

Edit Toolbar: This has features such as cut, copy, paste, trim audio, silence audio, undo, redo, sync-lock, zoom, etc.

Device Toolbar: This selects the audio host, recording device, recording channels and playback device.

- The Audio Host selects a specific interface, using which Audacity connects with the selected playback and recording devices. In Windows, MME is the default setting and is compatible with almost all audio devices, while on Linux the option is ALSA.
- The Recording Device can be chosen as per